NATIONAL UNIVERSITY OF TECHNOLOGY
(Department of Computer Science)
CS160– Database Systems
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Assignment 06

Total Marks: 10

Question [CLO: 2, PLO: 3]

A hospital consists of a number of specialized clinics (such as Maternity, Pediatrics, Cardiology, etc.). Each clinic has a number of doctors who work in it. Doctors are specialists in some branch of medicine and can be associated with at most one clinic of the hospital. Each clinic hosts a number of patients. On admission, the personal details of every patient are recorded. We also want to store information about medical tests undertaken by a patient, after recommendation of a specialist. A number of tests may be conducted for each patient. In addition, we would like to keep the following information:

• For each clinic, a name that uniquely identify it and the total number of rooms in the clinic. A clinic can have more than one doctor who works in it, can have more than one patient, and is managed by exactly one doctor.

• For each doctor, a license id which uniquely identifies him or her, the first and last name, the telephone number(s) (there can be more than one), the office, and the specialty. A doctor works in only one clinic. We want to keep, as well, the date when the doctor started working in the clinic.

• For each patient, a SIN (social insurance number) which uniquely identifies a patient, the first and last name, and the sex. A patient can undertake many medical tests and a medical test can be undertaken by many patients. For each patient hosted in a clinic, we want to keep, as well, the date when the patient arrived at the clinic and the date when they left the clinic.

• For each medical test, a test code that uniquely identifies it, a description, a cost, and the result of the test for each patient. The result of a test can be either negative or positive.

➢ *Determine* the main entities involved in this system and *design* their attributes and primary keys to ensure data integrity.

➢ *Determine* the relationships between these entities, specifying their connectivities and structural constraints.

➢ *Design* an ER Diagram (ERD) to represent this system, including entities, relationships, and necessary attributes.

➢ Evaluate any associative entities required for handling many-to-many relationships and explain their purpose.

➢ Translate your Entity-Relationship Model (ER Diagram) into Relational Schema.

ANSWER

Step 1: Identify Main Entities & Attributes

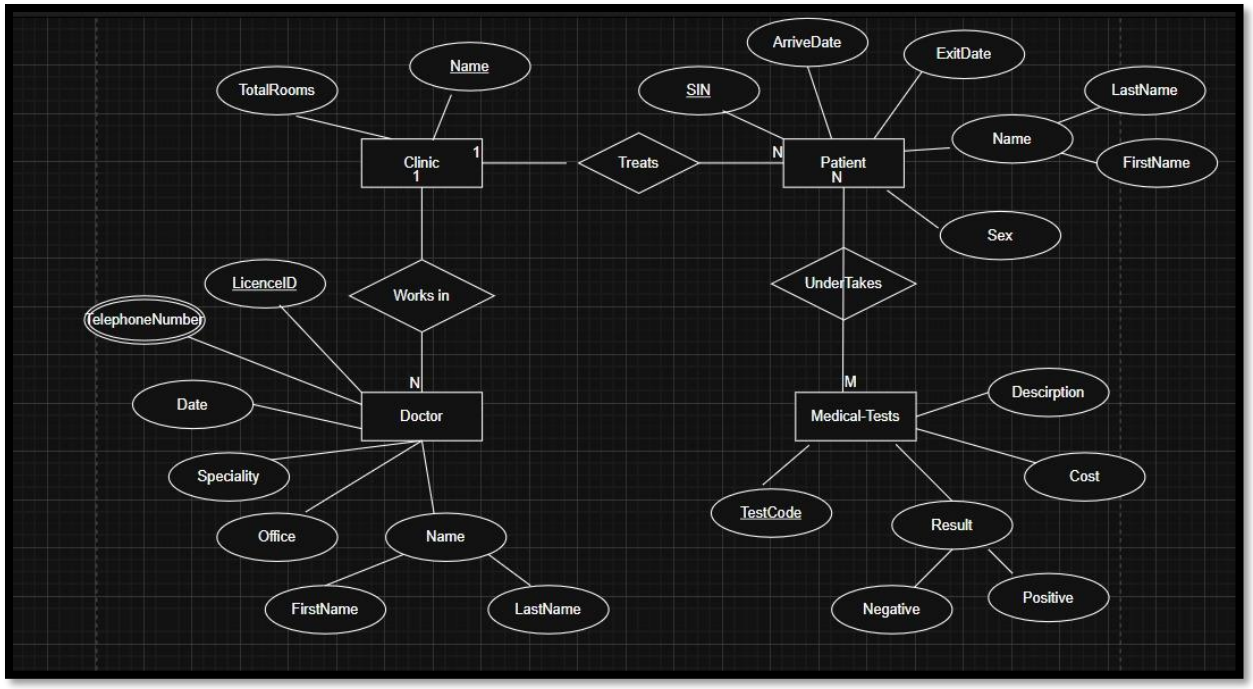
The core entities in this hospital system include:

1. Clinic - Attributes: Clinic_ID (PK), Name, Total_Rooms, Manager_Doctor_ID (FK).
2. Doctor - Attributes: Doctor_ID (PK), License_ID, First_Name, Last_Name, Telephone_Number(s), Office, Specialty, Clinic_ID (FK), Start_Date.
3. Patient - Attributes: Patient_ID (PK), SIN, First_Name, Last_Name, Sex, Arrival_Date, Departure_Date, Clinic_ID (FK).
4. Medical_Test - Attributes: Test_Code (PK), Description, Cost.
5. Test_Result (Associative Entity) - Attributes: Test Result ID (PK), Patient ID (FK), Test Code (FK), Result (Positive/Negative).

Step 2: Define Relationships AND Connectivities

1. Clinic - Doctor (1:M) → Each clinic employs multiple doctors, but a doctor works in only one clinic.
2. Clinic - Patient (1:M) → A clinic hosts multiple patients, but a patient belongs to only one clinic at a time.
3. Doctor – Medical Test (M:M via Test Result) → Multiple patients can undertake multiple tests, handled by different doctors.
4. Patient – Medical Test (M:M via Test Result) → A patient can undergo multiple medical tests, and a test can be undertaken by different patients.

Step 3: ER Diagram Design



Step 4: Evaluate any associative entities required for handling many-to-many relationships and explain their purpose.

Two associative entities are used:

1. **Patient_Test:** Resolves the many-to-many between Patient and MedicalTest.
    - Attributes: SIN, test_code, result.
2. **Patient_Clinic_Stay:** Enhances the Patient-Clinic relationship by storing the stay period.
    - Attributes: SIN, clinic_name, arrival_date, exit_date.

These entities help represent extra data about relationships, like test results and stay durations.

Step 5. Translate your Entity-Relationship Model (ER Diagram) into Relational Schema.

Relational Schema:

1. **Clinic:** (name [PRIMARY KEY], total_rooms).
2. **Doctor :** (license_id [PRIMARY KEY], f_name, l_name, office, specialty, start_date, clinic_name [FOREIGN KEY]).
3. **Doctor:** (license_id [PRIMARY KEY], telephone_number).
4. **ClinicHave:** (clinic_name PRIMARY KEY REFERENCES Clinic, manager_license_id UNIQUE REFERENCES Doctor)
5. **Patient:** (SIN [PRIMARY KEY] , f_name, l_name, sex,arrive-date, exit-date) MedicalTest(test_code PRIMARY KEY, description, cost, negative , positive)
6. **Patient-UnderTakes:** (SIN FOREIGN KEY REFERENCES Patient, test_code FOREIGN KEY REFERENCES MedicalTest, result, PRIMARY KEY(SIN, test_code))
7. **Patient_Treats:** (SIN FOREIGN KEY REFERENCES Patient, clinic_name FOREIGN KEY REFERENCES Clinic, arrival_date, exit_date, PRIMARY KEY(SIN, clinic_name, arrival_date))

These schemas reflect a normalized structure supporting data consistency and avoiding redundancy.